

Week 4: Daily Journals

Monday:

I began the week collaborating with Kevin to test the Software Defined Radio (SDR) integration within the Mode Manager program. We configured the antenna and SDRplay hardware inside the CubeSat laboratory. During initial execution, the script failed to output the expected .iq file. The remainder of the day was dedicated to debugging the software interface and verifying the data pipeline between the SDR hardware and the host script.

Tuesday:

I continued troubleshooting the SDR subsystem. Working alongside Kevin, we successfully resolved the file execution issue and achieved stable .IQ file generation. We spent the remainder of the day performing data validation on the output files to ensure the captured signal data was uncorrupted and legitimate.

Wednesday:

With the SDR module verified and producing accurate data, I pivoted to testing our two imaging payloads. I connected the 630 nm camera to the Raspberry Pi 4's USB 3.0 port and executed the capture script. The program generated an output file, but the resulting image was a solid green frame, indicating a data or hardware fault. I refactored the script to monitor system power metrics during the capture sequence; subsequent execution threw a system undervoltage warning.

Thursday:

To address the camera's power constraints, I attempted to capture images using a significantly lower frame rate to reduce transient power consumption. However, the system continued to trigger the undervoltage fault. To definitively isolate the root cause, I bridged a digital multimeter (DMM) into the hardware interface. The DMM confirmed a significant voltage drop underload, verifying that the Raspberry Pi 4's USB bus cannot supply sufficient current to power the camera directly.

Friday:

The first half of the day was spent exploring hardware workarounds to resolve the camera power issue, though a direct fix on the current setup was not achieved. To verify the camera hardware itself was fully functional, I tested it using an external laptop provided by Mr. Sell; the camera initialized and captured images successfully, confirming the issue is strictly related to the Pi's power bus limitations. I dedicated the remainder of the day to unboxing and staging the X-Ray sensor received from Chestnut Ridge High School in preparation for integration testing next week.

Pictures

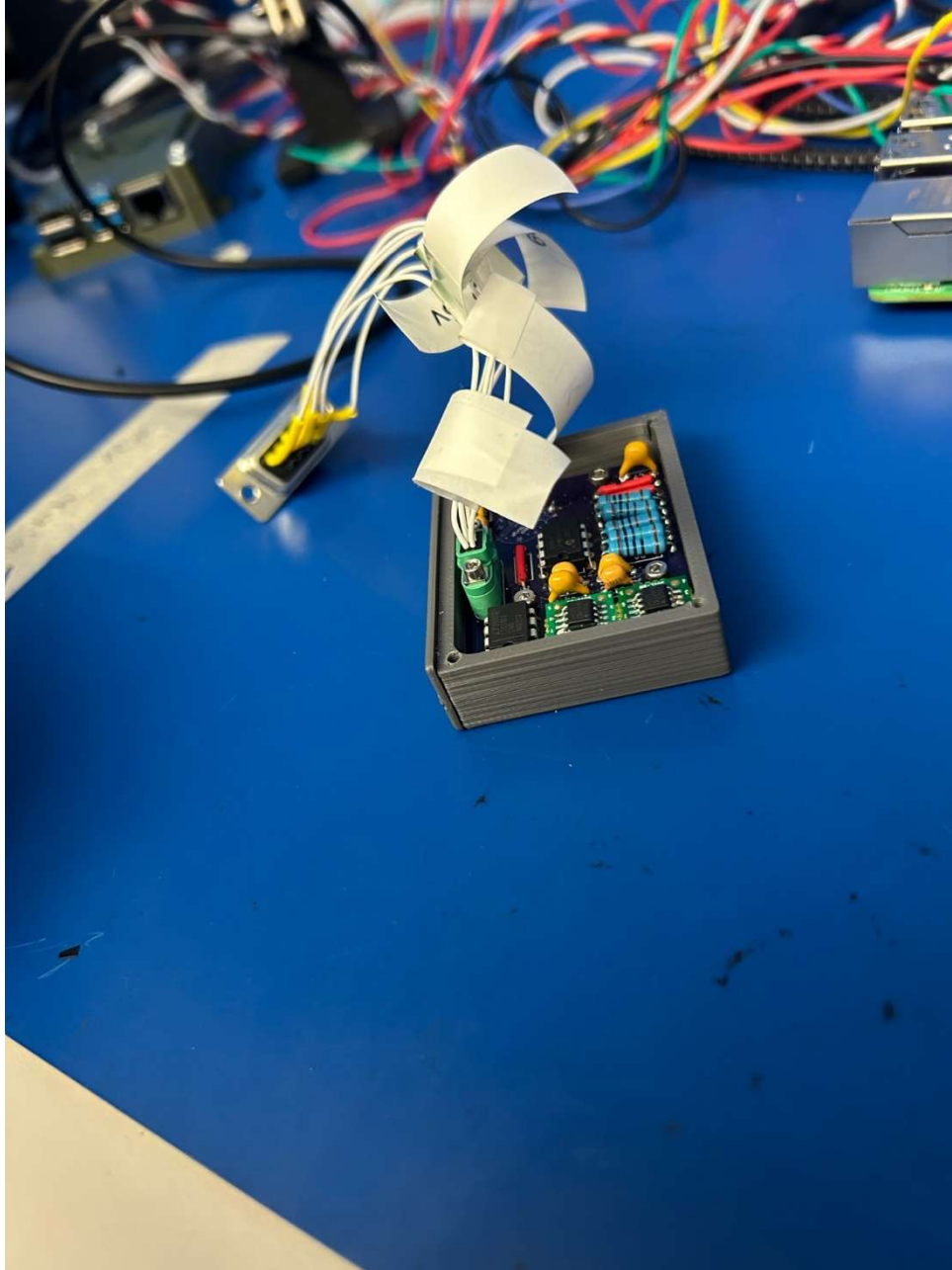


Figure 1: X-Ray Sensor